Liquidity in Automated Market Makers

Loss-versus-rebalancing: the LP's adverse selection without a book

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The question

An automated market maker (AMM) has no queue and no quotes: liquidity sits on a curve, and anyone can trade against it. **What does it cost to provide that liquidity, and how do fees, block time and concentration change the bill?**

- The pool's price lags the exchange price; arbitrageurs close the gap at the LP's expense.
- This is the same economics as stale quotes in a book (deck 8), with a closed form.

In this deck

1. Constant-product pools.
2. Loss-versus-rebalancing (LVR): derivation and exact mean.
3. Concentrated liquidity (derived here).
4. Fees and discrete blocks.

Constant-product pools

A pool holds x units of a risky asset and y units of the numeraire with $xy = L^2$; L is the liquidity. Its marginal price is $P = y/x$, so

$$x(P) = L/\sqrt{P}, \qquad y(P) = L\sqrt{P}, \qquad V(P) = xP + y = 2L\sqrt{P}.$$

The exchange price follows a geometric Brownian motion $\mathrm{d}P = \sigma P\,\mathrm{d}W$ with volatility σ (time$^{-1/2}$). Arbitrageurs trade the pool to P whenever it pays.

- The LP holds more of the asset after it falls and less after it rises: it is short volatility.
- "Impermanent loss" compares with buy-and-hold and mixes in market moves. LVR compares with a portfolio that holds the *same* $x(P_t)$ at all times, so it isolates the cost of trading at stale prices (Milionis et al., 2022).

Loss-versus-rebalancing

Rebalancing portfolio: holds $x(P_t)$ and trades at P_t, so $\mathrm{d}R = x(P)\,\mathrm{d}P$. The pool's value $V(P)$ satisfies $V'(P) = x(P)$ for any constant-function pool, because $y'(P) = -P\,x'(P)$. By Itô's formula

$$\mathrm{d}V - \mathrm{d}R = \tfrac{1}{2}V''(P)\,\sigma^2 P^2\,\mathrm{d}t = \tfrac{1}{2}x'(P)\,\sigma^2 P^2\,\mathrm{d}t \le 0.$$

So the LP loses at the rate

$$\frac{\mathrm{d}\,\mathrm{LVR}}{\mathrm{d}t} = \frac{\sigma^2 P^2}{2}\,|x'(P)| = \frac{\sigma^2}{8}\,V \quad \text{(constant product)}.$$

Exact mean. V is proportional to $\sqrt{P}$, and $\mathbb{E}\sqrt{P_T} = \sqrt{P_0}\,e^{-\sigma^2 T/8}$, so

$$\mathbb{E}[\mathrm{LVR}_T] = V_0\big(1 - e^{-\sigma^2 T/8}\big).$$

Units. σ^2 is per unit time, so $\sigma^2 V/8$ is value per unit time and $\sigma^2 T$ is dimensionless. Small T: $\mathbb{E}[\mathrm{LVR}_T] \approx \sigma^2 V_0 T/8$.

Concentrated liquidity

On a range $[P_a, P_b]$, $x(P) = L(P^{-1/2} - P_b^{-1/2})$ and $y(P) = L(P^{1/2} - P_a^{1/2})$ while P is inside. Then $x'(P) = -\frac{1}{2}LP^{-3/2}$, the same as for the full curve, so

$$\frac{\mathrm{d}\,\mathrm{LVR}}{\mathrm{d}t} = \frac{\sigma^2}{4}\,L\sqrt{P}, \qquad \text{while} \qquad V = 2L\sqrt{P} - LP\,P_b^{-1/2} - LP_a^{1/2}.$$

For a range $[P/w, Pw]$ centred in log price on the current P, with $w > 1$, $V = 2L\sqrt{P}(1 - w^{-1/2})$ and

$$\boxed{\frac{\mathrm{LVR}}{V} = \frac{\sigma^2}{8\,(1 - w^{-1/2})}}$$ per unit time, while P is in range.

- $w \to \infty$ recovers $\sigma^2/8$. A narrow range multiplies the cost per unit of capital by $1/(1 - w^{-1/2})$; it also multiplies fee income per unit of capital, which is why ranges are narrow.
- Derived here from the LVR rate of Milionis et al. (2022); checked by simulation to 2%.

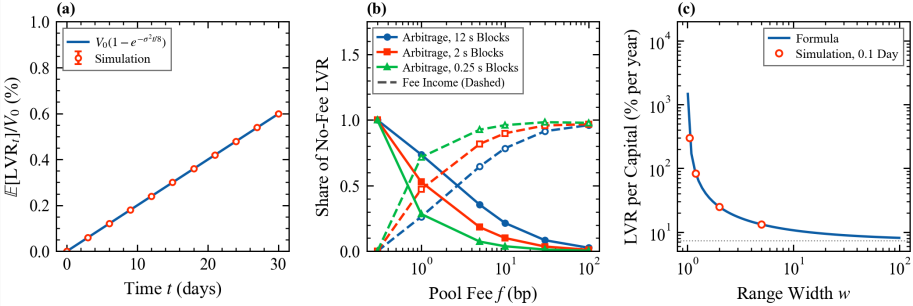
Fees and block time

With a proportional fee f, arbitrage pays only when the pool price is more than about f away from P: a **no-trade band**. With arbitrage only at block times Δt apart, the price can jump through the band between blocks.

- Arbitrage profit falls below LVR; part of LVR comes back to the LP as fees (Milionis et al., 2023).
- The split depends on how far the price moves in one block relative to the fee, $\sigma\sqrt{\Delta t}$ against f.

Simulation here. The exchange price moves as a geometric Brownian motion; at each block the arbitrageur trades the pool to the edge of the band if it is outside, paying fee f on the input. Arbitrage profit and fees are recorded per path and divided by the no-fee LVR.

LVR: fees, block time and concentration



(a) $\sigma = 0.04$ day$^{-1/2}$, about 76% a year: LVR costs 7.3% of capital per year; the 30-day simulation gives $0.598 \pm 0.001\%$ against the exact 0.598%.

(b) At 5 bp and 12 s blocks, arbitrage keeps 0.35 and fees return 0.65 of no-fee LVR; at 30 bp, 0.08 and 0.92. (c) A $w = 1.1$ range pays 21.5× the full-range LVR per unit capital.

Discussion and takeaways

Book LP and AMM LP, side by side

- Book: stale quotes picked off in latency races; cost about a third of the effective spread on the LSE (Aquilina et al., 2022).
- AMM: stale pool price arbitraged every block; cost $\sigma^2/8$ of capital per unit time.
- Both are adverse selection from public information, not from private information.

Assumptions. Geometric Brownian motion, no gas cost, a single competitive arbitrageur, no noise traders in the simulation.

For a liquidity provider

1. Price the position by LVR, not by impermanent loss.
2. Fees must exceed the arbitrage share of LVR; faster blocks help the LP.
3. Concentration scales cost and income together; it pays only if fee income per unit of capital scales at least as fast.

Open question. Which design, book or AMM, fee level and block time, minimizes the total cost of liquidity for a given asset.

Reproduce

Code: `amm.py`. Experiment: `e13_lvr.py`.

```
from stochastic_lob import amm
amm.expected_lvr_cpmm(T=365, sigma=0.04)              # value; V_0 = 2 here
amm.lvr_per_capital_concentrated(sigma=0.04, w=1.1)    # per day, in range
```

References i

Matteo Aquilina, Eric Budish, and Peter O'Neill. Quantifying the high-frequency trading "arms race". *The Quarterly Journal of Economics*, 137(1):493–564, 2022.

Jason Milionis, Ciamac C. Moallemi, Tim Roughgarden, and Anthony Lee Zhang. Automated market making and loss-versus-rebalancing, 2022. arXiv:2208.06046.

Jason Milionis, Ciamac C. Moallemi, and Tim Roughgarden. Automated market making and arbitrage profits in the presence of fees, 2023. arXiv:2305.14604.